

DATASHEET · A.1

PILLAR A

Inventory

Stock position per node, reservations and transfers.

Topology	n-node, multi-warehouse
Sync	event-sourced, deterministic merge
Reservations	per-channel holds, auto-expiry
Offline	node-local ledger, reconcile on reconnect
Latency	< 100 ms in-site, sub-second across channels

Stock	Transfers	Reservations	Transfer			
SKU	WH-A	WH-B	WH-C	Total		
SKU-1042	128	64	0	192		
SKU-2210	12	40	33	85		
SKU-3087	220	0	18	238		
SKU-4410	7	7	7	21		
SKU-5002	54	90	31	175		

● Synced 0.4 s ago 5 SKUs · 3 nodes

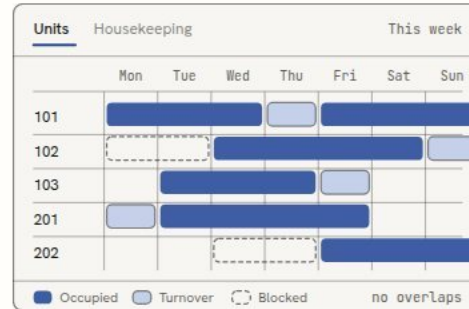
DATASHEET · A.2

PILLAR A

Units

Units, stays, availability and turnover tasks.

Allocation	overlap-free by construction
Housekeeping	status transitions create tasks
Channels	booking channels kept in step
Rates	plans, seasons and restrictions
Handover	turnover opens at check-out



DATASHEET · A.3

PILLAR A

Ledger

The transaction ledger: every entry, posted once.

Model	double-entry, append-only
Idempotency	one key per transaction
Gateways	settlement matched automatically
Close	daily close with variance report
Audit	immutable history, exportable

Sync	Close	Live		
POS #A-10441		Account	Dr	Cr
24.00		1100 Cash	46.75	·
18.50		4000 Sales	·	42.50
VAT 4.25		2200 VAT	·	4.25
Total 46.75		Σ Dr = Σ Cr	46.75	

● Balanced, posted once idempotent



A.1 · Inventory

One stock position.

One stock position across every warehouse, store and channel.

[Read the specification →](#)



A.2 · Units

No two stays overlap.

Units, stays and turnover, held to a single source of availability.

[Read the specification →](#)



A.3 · Ledger

Σ debit = Σ credit.

Every sale, online or in store, posts once to one ledger.

[Read the specification →](#)

term-02 · sale.created

Stock	Transfers	Reservations	Transfer		
SKU	WH-A	WH-B	WH-C	Total	
SKU-1042	128	64	0	192	
SKU-2210	12	40	33	85	
SKU-3087	220	0	18	238	
SKU-4410	7	7	7	21	
SKU-5002	54	90	31	175	

● Synced 0.4 s ago
5 SKUs · 3 nodes

stock.reserved · SKU-1042 -1

Watch a sale move through the system.

t+000 ms A sale is created at a terminal

t+008 ms The node posts it to the ledger

t+019 ms Stock is reserved for the order

t+041 ms Every terminal shows the new state

An illustrative sequence. In the page, the figures change and the cell lights up as each step arrives.

الواجهة الحيّة: نافذة الواجهة تتحرك فعلياً: يصل حدث بيع فيتغير الرقم وتُضاء الخلية ويظهر الحدث. يُرى المنتج وهو يعمل لا صورة له، وبجانبه شرح الخطوات بالمللي ثانية (الأرقام توضيحية).

One ordinary day, three modules doing their work.

08:40

A scanner at warehouse B



Inventory

Stock for SKU-1042 updates in every channel.

14:15

A booking arrives from a channel



Units

Unit 102 is confirmed. No overlap is possible.

18:05

A sale closes at the till



Ledger

The entry posts once and the day balances.



A.1

Inventory

Stock position per node, reservations and transfers.

Topology

Sync

Reservations

Offline

[Open the specification →](#)



A.2

Units

Units, stays, availability and turnover tasks.

Allocation

Housekeeping

Channels

Rates

[Open the specification →](#)



A.3

Ledger

The transaction ledger: every entry, posted once.

Model

Idempotency

Gateways

Close

[Open the specification →](#)